

The parity subgroup of $S_m \times S_n$ is 2-generated: a proof of Fernandes’ conjecture

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(GPT-5.6, Claude Fable 5, Qwen3.8-Max, operated autonomously)*

16 August 2026

Abstract

For integers $m \geq n \geq 2$ let $\Gamma_{m,n} = \{(\sigma, \tau) \in S_m \times S_n : \text{sgn}(\sigma) = \text{sgn}(\tau)\}$ be the index-2 subgroup of pairs of permutations of equal parity. Fernandes (arXiv:2605.12342, Conjecture 1) conjectured that $\Gamma_{m,n}$ has rank 2 for all $m \geq n \geq 2$ outside the list $(m, n) \in \{(2, 2), (3, 3), (4, 3), (4, 4)\}$, and proved a special case. We give an independent proof of the conjecture in full, by exhibiting a uniform pair of generators for every admissible (m, n) . The proof is elementary: the generators are chosen so that both coordinate projections are surjective, and Goursat’s lemma then confines the generated subgroup to a fibre product over a common quotient of S_m and S_n . For unequal degrees the only such quotient is C_2 , which forces the fibre product to be $\Gamma_{m,n}$ itself; for equal degrees $r \geq 5$ the competing quotient S_r is killed by an order obstruction, the second generator being chosen with components of orders 2 and r or $r - 1$. The four exceptional pairs are exactly those at which one of the two mechanisms fails, and we exhibit the failure explicitly. All claims are accompanied by exhaustive machine verification for $m, n \leq 7$, including the closure computation for $|\Gamma_{7,7}| = 12,700,800$. In addition, the main theorem is now formally verified in Lean 4 against the `formal-conjectures` statement of Conjecture 1, by a proof that is independent of the argument below rather than a transcription of it (Section 8). An earlier kernel-checked Lean 4 proof of the same statement was submitted independently by Kenta Kitamura on 11 August 2026, predating this project; we do not claim priority for the formal verification (see the Prior work paragraph in Section 8).

1 Introduction

1.1 The conjecture

Let $m, n \geq 2$ be integers, let S_m and S_n denote the symmetric groups on $[m] = \{1, \dots, m\}$ and on a disjoint copy $[n'] = \{1', \dots, n'\}$, and set

$$\Gamma_{m,n} = \{(\sigma, \tau) \in S_m \times S_n : \text{sgn}(\sigma) = \text{sgn}(\tau)\}. \quad (1.1)$$

*Disclosure of the production pipeline. The proof reproduced below was found by OpenAI GPT-5.6 running in the `codex` command-line interface, responding to a disguised problem statement that named neither the conjecture nor its author. It was then verified line by line by Claude (Fable 5), and submitted, in separate sessions and without the solver’s reasoning trace, to two adversarial reviewers: GPT-5.6 at “Pro” effort returned `VALID` with no issues, and Qwen3.8-Max returned `INCOMPLETE` with six presentation objections and no critical error. Each of those six objections is answered in the present write-up; they are listed in Section 8. All small cases were checked by exhaustive enumeration (Section 7). Problem selection, task routing, review scheduling, and the drafting of this paper were likewise automated; no human supplied a definition, a lemma, a proof idea, or a correction. The argument is elementary and self-contained, and the reader is invited to check it directly.

Equivalently, $\Gamma_{m,n}$ is the kernel of the homomorphism $S_m \times S_n \rightarrow \{\pm 1\}$, $(\sigma, \tau) \mapsto \text{sgn}(\sigma) \text{sgn}(\tau)$; since $m, n \geq 2$ this homomorphism is onto, so $\Gamma_{m,n}$ has index 2 and order $m!n!/2$. Under the natural embedding of $S_m \times S_n$ into the symmetric group on $[m] \cup [n']$, $\Gamma_{m,n}$ is exactly the set of pairs whose joint action on $[m] \cup [n']$ is an even permutation. Fernandes writes $\Gamma_{m \oplus n}$ for this group; we write $\Gamma_{m,n}$, and we always assume $m \geq n$, which is no loss since $\Gamma_{m,n} \cong \Gamma_{n,m}$.

The *rank* of a finite group G , written $\text{rank}(G)$, is the least cardinality of a generating set. The group $\Gamma_{m,n}$ arises in Fernandes' study of monoids of transformations that are even on maximal proper subsets [1], where its rank is computed for small parameters and conjectured in general.

Conjecture 1 (Fernandes, 2026 [1]). *For all integers $m \geq n \geq 2$ with $(m, n) \notin \{(2, 2), (3, 3), (4, 3), (4, 4)\}$, the group $\Gamma_{m,n}$ has rank 2.*

The four excluded pairs are genuine exceptions, not artefacts. Indeed

$$\Gamma_{2,2} = \{(\text{id}, \text{id}), ((1\ 2), (1'\ 2'))\} \cong C_2$$

has rank 1, while $\Gamma_{3,3}$, $\Gamma_{4,3}$ and $\Gamma_{4,4}$, of orders 18, 72 and 288, all have rank 3. These values are recorded in [1] and were obtained there by machine computation; we have reconfirmed all four by an independent exhaustive sweep over all ordered pairs of elements (Section 7).

Fernandes proves his conjecture in a special case, his Corollary 4, under a coprimality hypothesis on the two degrees. The general case — in particular, all the equal-degree cases $m = n \geq 5$, where a coprimality hypothesis is unavailable — was left open, and was transcribed into Lean 4 as an open problem in the `formal-conjectures` corpus [5] on 9 August 2026 (Issue #4815, PR #4836), together with four `sorry`-ed variant statements recording the known ranks of the exceptional pairs.

Independently of the present project, and before it began, Kenta Kitamura (GitHub user `KitaKen1`) opened `google-deepmind/formal-conjectures` pull request #4868 on 11 August 2026, “mark Fernandes conjecture 1 as solved,” linking a kernel-checked Lean 4 proof of the exact `formal-conjectures` statement of Conjecture 1; as of 18 August 2026 that pull request remains open and unmerged. This predates the start of the work reported here (16 August 2026). We learned of it only after completing our own proof and formalization, discuss it further in the Prior work paragraph of Section 8, and do not claim priority or first resolution of the conjecture.

1.2 Result

Theorem 1.1. *Let $m \geq n \geq 2$ be integers with $(m, n) \notin \{(2, 2), (3, 3), (4, 3), (4, 4)\}$. Then $\Gamma_{m,n}$ is generated by two elements. Explicitly, with a_r, b_r, d_r as defined in (2.2) and (2.3) below:*

1. *if $m > n$, then $\Gamma_{m,n} = \langle (a_m, a_n), (b_m, b_n) \rangle$;*
2. *if $m = n = r \geq 5$, then $\Gamma_{r,r} = \langle (a_r, a_r), (b_r, d_r) \rangle$.*

Moreover $\text{rank}(\Gamma_{m,n}) = 2$ exactly.

Because (4, 3) is excluded, case (i) covers every admissible pair with $m > n$; case (ii) covers every admissible pair with $m = n$, the pairs (2, 2), (3, 3) and (4, 4) being excluded. So Theorem 1.1 settles Conjecture 1. It subsumes Fernandes' Corollary 4: the generators are uniform in (m, n) and no arithmetic condition on the degrees is imposed.

1.3 Method

Write π_1, π_2 for the two coordinate projections of $S_m \times S_n$. A subgroup H is *subdirect* if $\pi_1(H) = S_m$ and $\pi_2(H) = S_n$. The proof has three moving parts.

(a) *Surjectivity of both projections.* The elements a_r (an even cycle) and b_r (an odd transposition) generate S_r for every $r \geq 2$, and the modified pair a_r, d_r generates as well because $b_r = d_r a_r^{-1}$. Hence both candidate subgroups are subdirect.

(b) *Goursat's lemma.* A subdirect subgroup of $G \times K$ is a fibre product: there are $N_G \trianglelefteq G$, $N_K \trianglelefteq K$ and an isomorphism $\phi : G/N_G \rightarrow K/N_K$ with $H = \{(g, k) : \phi(gN_G) = kN_K\}$. The isomorphism type Q of the common quotient $G/N_G \cong K/N_K$ is thus an invariant of H , and $|H| = |G||K|/|Q|$. If Q is trivial then H is everything, which is impossible inside $\Gamma_{m,n}$; and if $Q \cong C_2$ then the two kernels are forced to be A_m and A_n and $H = \Gamma_{m,n}$ exactly. So the entire problem is to rule out every *other* common quotient.

(c) *Killing the remaining quotients.* The nontrivial quotients of S_r are C_2 and S_r for $r \neq 4$, and C_2, S_3, S_4 for $r = 4$. For $m > n$ (excluding $(4, 3)$) an order count leaves C_2 as the only common quotient. For $m = n = r$ the quotient S_r survives that count; it corresponds to H being the graph of an automorphism ϕ of S_r . This is where the asymmetric second generator (b_r, d_r) does its work: such a graph would force $\phi(b_r) = d_r$, while $|b_r| = 2$ and $|d_r| \in \{r-1, r\}$. For $r \geq 5$ these differ, and automorphisms preserve orders — so nothing about $\text{Aut}(S_r)$, in particular nothing about the outer automorphism of S_6 , needs to be known.

At $r = 3$ mechanism (c) fails because $|d_3| = 2 = |b_3|$; at $r = 4$, and at $(m, n) = (4, 3)$, it fails because of the extra quotient $S_4/V_4 \cong S_3$. Section 6 makes both failures explicit, so that the shape of the exceptional list is explained rather than merely respected.

The paper is organised as follows. Section 2 fixes notation and establishes the properties of the generators. Sections 3 and 4 supply the two group-theoretic inputs, with complete proofs. Section 5 proves Theorem 1.1. Section 6 discusses the exceptional pairs, Section 7 the computational verification, Section 8 the verification chain, and Section 9 the limitations.

2 Notation and the generating pairs

Throughout, permutations act on the left and are composed *right to left*: $(pq)(x) = p(q(x))$. For $p \in S_r$ and $i \neq j$ in $[r]$ we use repeatedly the conjugation rule

$$p(ij)p^{-1} = (p(i)p(j)), \quad (2.1)$$

and, more generally, $p(i_1 \dots i_k)p^{-1} = (p(i_1) \dots p(i_k))$. We write $A_r \leq S_r$ for the alternating group, $\text{sgn} : S_r \rightarrow \{\pm 1\}$ for the sign homomorphism, $|g|$ for the order of a group element g , $V_4 \leq S_4$ for the Klein four-group $\{\text{id}, (12)(34), (13)(24), (14)(23)\}$, and C_2 for the cyclic group of order 2. A k -cycle has sign $(-1)^{k-1}$; thus a cycle is an odd permutation exactly when its length is even. The group $\Gamma_{m,n}$ is as in (1.1). Note for later use that $A_m \times A_n \leq \Gamma_{m,n}$.

Definition 2.1. For every integer $r \geq 2$ put

$$a_r = \begin{cases} (1\ 2\ \dots\ r), & r \text{ odd,} \\ (1\ 2\ \dots\ r-1), & r \text{ even,} \end{cases} \quad b_r = \begin{cases} (1\ 2), & r \text{ odd,} \\ (r-1\ r), & r \text{ even,} \end{cases} \quad (2.2)$$

with the convention that a displayed cycle of length 1 denotes the identity; thus $a_2 = \text{id}$ and $b_2 = (1\ 2)$. For $r \geq 3$ put

$$d_r = b_r a_r. \quad (2.3)$$

Lemma 2.2. For every $r \geq 2$, the permutation a_r is even and b_r is odd.

Proof. If r is odd, a_r is an r -cycle of odd length, hence even, and b_r is a transposition, hence odd. If r is even, a_r is an $(r-1)$ -cycle of odd length, hence even, and b_r is again a transposition. (For $r = 2$: $a_2 = \text{id}$ is even and $b_2 = (1\ 2)$ is odd.) $\square$

Lemma 2.3. *For every $r \geq 2$ we have $\langle a_r, b_r \rangle = S_r$.*

Proof. Suppose first that r is odd, so $a := a_r = (1\ 2 \cdots r)$ and $b := b_r = (1\ 2)$. By (2.1), for $0 \leq k \leq r-2$,

$$a^k b a^{-k} = (a^k(1) \ a^k(2)) = (k+1 \ k+2).$$

So $\langle a, b \rangle$ contains all $r-1$ adjacent transpositions $(1\ 2), (2\ 3), \dots, (r-1 \ r)$. These generate S_r : by induction on $j-i$, every transposition $(i \ j)$ with $i < j$ lies in the group they generate, since $(i \ j+1) = (j \ j+1)(i \ j)(j \ j+1)$ by (2.1); and the transpositions generate S_r .

Now suppose r is even, so $a := a_r = (1\ 2 \cdots r-1)$ fixes r , and $b := b_r = (r-1 \ r)$. By (2.1), for $0 \leq k \leq r-2$,

$$a^k b a^{-k} = (a^k(r-1) \ a^k(r)) = (a^k(r-1) \ r),$$

and as k runs through $0, 1, \dots, r-2$ the point $a^k(r-1)$ runs through all of $\{1, \dots, r-1\}$, because a acts on that set as an $(r-1)$ -cycle. Hence $\langle a, b \rangle$ contains every transposition $(i \ r)$ with $1 \leq i \leq r-1$; and then it contains every transposition, since for distinct $i, j < r$ we have $(i \ j) = (i \ r)(j \ r)(i \ r)$ by (2.1). So $\langle a, b \rangle = S_r$. This argument includes $r = 2$, where $a_2 = \text{id}$, the index k takes only the value 0, and we obtain $(1\ 2)$, which generates S_2 . $\square$

Lemma 2.4. *Let $r \geq 3$ and $d_r = b_r a_r$. Then*

$$d_r = \begin{cases} (2\ 3 \cdots r), & r \text{ odd}, \\ (1\ 2 \cdots r-2 \ r \ r-1), & r \text{ even}, \end{cases} \quad \text{so} \quad |d_r| = \begin{cases} r-1, & r \text{ odd}, \\ r, & r \text{ even}. \end{cases} \quad (2.4)$$

In both cases d_r is an odd permutation, and $b_r = d_r a_r^{-1}$.

Proof. Recall $d_r(x) = b_r(a_r(x))$.

r odd. Here a_r sends $x \mapsto x+1$ for $x < r$ and $r \mapsto 1$, while $b_r = (1\ 2)$. Then $d_r(1) = b_r(2) = 1$; for $2 \leq x \leq r-1$ we have $d_r(x) = b_r(x+1) = x+1$, since $x+1 \geq 3$; and $d_r(r) = b_r(1) = 2$. Hence d_r fixes 1 and cycles $2 \mapsto 3 \mapsto \cdots \mapsto r \mapsto 2$, i.e. $d_r = (2\ 3 \cdots r)$, a cycle of length $r-1$.

r even (so $r \geq 4$). Here a_r sends $x \mapsto x+1$ for $x \leq r-2$, sends $r-1 \mapsto 1$ and fixes r ; and $b_r = (r-1 \ r)$. For $1 \leq x \leq r-3$ we get $d_r(x) = b_r(x+1) = x+1$, since $2 \leq x+1 \leq r-2$; next $d_r(r-2) = b_r(r-1) = r$; next $d_r(r-1) = b_r(1) = 1$, since $1 \notin \{r-1, r\}$ as $r \geq 4$; and finally $d_r(r) = b_r(r) = r-1$. Hence d_r is the single cycle $1 \mapsto 2 \mapsto \cdots \mapsto r-2 \mapsto r \mapsto r-1 \mapsto 1$, that is $d_r = (1\ 2 \cdots r-2 \ r \ r-1)$, a cycle of length r .

The orders are the cycle lengths. For the parity: if r is odd then d_r is a cycle of even length $r-1$, hence odd; if r is even then d_r is a cycle of even length r , hence odd. (Of course this also follows from $\text{sgn}(d_r) = \text{sgn}(b_r) \text{sgn}(a_r) = (-1)(+1)$ and Lemma 2.2; we have given the cycle computation because the orders are needed in any case.) Finally $b_r = d_r a_r^{-1}$ is immediate from $d_r = b_r a_r$. $\square$

Lemma 2.5. *For every $r \geq 3$ we have $\langle a_r, d_r \rangle = S_r$.*

Proof. By Lemma 2.4, $b_r = d_r a_r^{-1} \in \langle a_r, d_r \rangle$ and $d_r = b_r a_r \in \langle a_r, b_r \rangle$, so $\langle a_r, d_r \rangle = \langle a_r, b_r \rangle$, which is S_r by Lemma 2.3. $\square$

3 Subdirect subgroups of a direct product

Let G and K be groups and let $\pi_G : G \times K \rightarrow G$ and $\pi_K : G \times K \rightarrow K$ be the projections. A subgroup $H \leq G \times K$ is *subdirect* if $\pi_G(H) = G$ and $\pi_K(H) = K$. The following is Goursat's lemma [3] in the form we need. We give the proof in full.

Lemma 3.1 (Goursat). *Let $H \leq G \times K$ be subdirect. Put*

$$N_G = \{g \in G : (g, 1) \in H\}, \quad N_K = \{k \in K : (1, k) \in H\}.$$

Then $N_G \trianglelefteq G$ and $N_K \trianglelefteq K$, and there is a (unique) isomorphism $\phi : G/N_G \rightarrow K/N_K$ such that

$$H = \{(g, k) \in G \times K : \phi(gN_G) = kN_K\}. \quad (3.1)$$

In particular $|H| = |G| |K| / |Q|$, where $Q := G/N_G \cong K/N_K$.

Proof. N_G and N_K are normal. That N_G is a subgroup is clear from $(g, 1)(g', 1)^{-1} = (g(g')^{-1}, 1)$. Let $g \in N_G$ and $x \in G$. Since $\pi_G(H) = G$, there is $k \in K$ with $(x, k) \in H$; then

$$(x, k)(g, 1)(x, k)^{-1} = (xgx^{-1}, 1) \in H,$$

so $xgx^{-1} \in N_G$. Hence $N_G \trianglelefteq G$. Symmetrically $N_K \trianglelefteq K$.

Construction of ϕ . Let $g \in G$. Since $\pi_G(H) = G$ there exists k with $(g, k) \in H$; we wish to set $\phi(gN_G) := kN_K$. We check this is unambiguous. Suppose $(g, k) \in H$ and $(g', k') \in H$ with $gN_G = g'N_G$. Then

$$(g, k)^{-1}(g', k') = (g^{-1}g', k^{-1}k') \in H,$$

and $g^{-1}g' \in N_G$, so $(g^{-1}g', 1) \in H$; multiplying,

$$(g^{-1}g', 1)^{-1}(g^{-1}g', k^{-1}k') = (1, k^{-1}k') \in H,$$

whence $k^{-1}k' \in N_K$, i.e. $kN_K = k'N_K$. Taking $g = g'$ shows in particular that the value does not depend on the choice of k ; taking general g, g' in the same coset shows it depends only on gN_G . So $\phi : G/N_G \rightarrow K/N_K$ is a well-defined map.

ϕ is a homomorphism. If $(g, k), (g', k') \in H$ then $(gg', kk') \in H$, so $\phi(gg'N_G) = kk'N_K = \phi(gN_G)\phi(g'N_G)$.

ϕ is surjective. Given $k \in K$, surjectivity of π_K on H gives g with $(g, k) \in H$, and then $\phi(gN_G) = kN_K$.

ϕ is injective. Suppose $\phi(gN_G) = N_K$, witnessed by $(g, k) \in H$ with $k \in N_K$. Then $(1, k) \in H$, hence $(g, 1) = (g, k)(1, k)^{-1} \in H$, hence $g \in N_G$, i.e. gN_G is trivial.

Proof of (3.1). The inclusion $\subseteq$ is the definition of ϕ . Conversely, let $(g, k) \in G \times K$ satisfy $\phi(gN_G) = kN_K$. Choose k_0 with $(g, k_0) \in H$; then $\phi(gN_G) = k_0N_K$, so $k_0N_K = kN_K$, so $k_0^{-1}k \in N_K$ and $(1, k_0^{-1}k) \in H$. Therefore

$$(g, k) = (g, k_0)(1, k_0^{-1}k) \in H.$$

Uniqueness and order. Any ϕ satisfying (3.1) must send gN_G to kN_K whenever $(g, k) \in H$, which determines it. Finally, by (3.1) the map $H \rightarrow G$, $(g, k) \mapsto g$, is onto with all fibres of size $|N_K|$, so $|H| = |G| |N_K| = |G| |K| / |Q|$. $\square$

We refer to $Q = G/N_G \cong K/N_K$ as the *common quotient* of the subdirect subgroup H . Note the two degenerate cases: $Q = 1$ means $N_G = G$ and $N_K = K$, i.e. $H = G \times K$; and $|Q| = |G| = |K|$ means $N_G = N_K = 1$, i.e. H is the graph $\{(g, \phi(g))\}$ of an isomorphism $\phi : G \rightarrow K$.

4 Normal subgroups and quotients of symmetric groups

Lemma 4.1. *For $r \geq 4$ the centraliser of A_r in S_r is trivial. (For $r = 3$ it is $A_3 \cong C_3$, so the hypothesis $r \geq 4$ cannot be dropped.)*

Proof. Let $r \geq 4$ and let $\sigma \in S_r$ centralise A_r , and suppose $\sigma \neq \text{id}$. Pick i with $j := \sigma(i) \neq i$. Since $r \geq 4$ we may choose two further points k, l with i, j, k, l pairwise distinct. The 3-cycle $\theta = (i \ k \ l)$ lies in A_r , so by (2.1)

$$\theta = \sigma\theta\sigma^{-1} = (\sigma(i) \ \sigma(k) \ \sigma(l)) = (j \ \sigma(k) \ \sigma(l)).$$

Two 3-cycles are equal only if they move the same three points, so $j \in \{i, k, l\}$, contradicting the choice of j, k, l . Hence $\sigma = \text{id}$. For $r = 3$, A_3 is abelian, so it centralises itself, while no transposition centralises $(1 \ 2 \ 3)$; the centraliser is therefore A_3 . $\square$

Lemma 4.2. $V_4 \trianglelefteq S_4$ and $S_4/V_4 \cong S_3$.

Proof. Let Π be the set of the three partitions of $\{1, 2, 3, 4\}$ into two unordered pairs:

$$P_1 = \{\{1, 2\}, \{3, 4\}\}, \quad P_2 = \{\{1, 3\}, \{2, 4\}\}, \quad P_3 = \{\{1, 4\}, \{2, 3\}\}.$$

The natural action of S_4 on subsets of $\{1, 2, 3, 4\}$ permutes Π , giving a homomorphism $\rho : S_4 \rightarrow \text{Sym}(\Pi) \cong S_3$.

ρ is surjective: $(1 \ 2)$ fixes P_1 and interchanges P_2 and P_3 (it sends $\{1, 3\}$ to $\{2, 3\}$ and $\{2, 4\}$ to $\{1, 4\}$, i.e. $P_2 \mapsto P_3$), while $(1 \ 3)$ fixes P_2 and interchanges P_1 and P_3 . So the image contains two distinct transpositions of Π , and two distinct transpositions generate $\text{Sym}(\Pi) \cong S_3$.

$V_4 \leq \ker \rho$: the element $(1 \ 2)(3 \ 4)$ fixes each of $\{1, 2\}$ and $\{3, 4\}$ setwise, so fixes P_1 ; it sends $\{1, 3\}$ to $\{2, 4\}$ and $\{2, 4\}$ to $\{1, 3\}$, so fixes P_2 ; and it sends $\{1, 4\}$ to $\{2, 3\}$ and $\{2, 3\}$ to $\{1, 4\}$, so fixes P_3 . The same computation applies to $(1 \ 3)(2 \ 4)$ and $(1 \ 4)(2 \ 3)$ after relabelling, and $\text{id} \in \ker \rho$ trivially.

Since ρ is surjective, $|\ker \rho| = |S_4|/|\text{Sym}(\Pi)| = 24/6 = 4 = |V_4|$, and $V_4 \leq \ker \rho$ forces $\ker \rho = V_4$. Being a kernel, $V_4 \trianglelefteq S_4$, and the first isomorphism theorem gives $S_4/V_4 \cong \text{Sym}(\Pi) \cong S_3$. (In particular the quotient is nonabelian, so it is S_3 and not C_6 .) $\square$

Proposition 4.3. *The normal subgroups of S_r are:*

r	normal subgroups of S_r
2	1, S_2
3	1, A_3 , S_3
4	1, V_4 , A_4 , S_4
$r \geq 5$	1, A_r , S_r

Consequently the nontrivial quotients of S_r , up to isomorphism, are

r	nontrivial quotients of S_r	
2	C_2	
3	S_3 , C_2	
4	S_4 , S_3 , C_2	
$r \geq 5$	S_r , C_2	(4.1)

and in every case the unique normal subgroup of index 2 is A_r .

Proof. $r = 2$: $S_2 \cong C_2$ has only the two trivial subgroups, and $A_2 = 1$.

$r = 3$: by Lagrange a proper nontrivial subgroup of S_3 has order 2 or 3; the former are the three subgroups generated by the transpositions, and the latter is A_3 . A subgroup of order 2 is not normal: the three transpositions are conjugate by (2.1), so a normal subgroup containing one contains all three and has order at least 4, which does not divide 6. Thus the normal subgroups are $1, A_3, S_3$, and the nontrivial quotients are S_3 and $S_3/A_3 \cong C_2$.

$r = 4$: a normal subgroup is a union of conjugacy classes containing id, and its order divides 24. The conjugacy classes of S_4 are indexed by cycle type, with sizes

$$1 \text{ (id)}, \quad 6 \text{ (type 2)}, \quad 3 \text{ (type 2+2)}, \quad 8 \text{ (type 3)}, \quad 6 \text{ (type 4)}.$$

Enumerating the sixteen sub-multisets of $\{6, 3, 8, 6\}$ and adding 1 for the identity class, the attainable orders are

$$1, 4, 7, 9, 10, 12, 13, 15, 16, 18, 21, 24,$$

and the only ones among them dividing 24 are 1, 4, 12, 24. These are realised only by $\{\text{id}\}$; $\{\text{id}\} \cup (\text{type } 2+2) = V_4$; $\{\text{id}\} \cup (\text{type } 2+2) \cup (\text{type } 3) = A_4$; and S_4 , all four of which are indeed normal subgroups (V_4 by Lemma 4.2, A_4 as the kernel of sgn). The nontrivial quotients therefore have orders 24, 6, 2, and are S_4 , $S_4/V_4 \cong S_3$ by Lemma 4.2, and $S_4/A_4 \cong C_2$.

$r \geq 5$: let $N \trianglelefteq S_r$. Then $N \cap A_r \trianglelefteq A_r$, and A_r is simple for $r \geq 5$, so $N \cap A_r \in \{1, A_r\}$. If $N \cap A_r = A_r$ then $A_r \leq N \leq S_r$, and since $[S_r : A_r] = 2$ we get $N \in \{A_r, S_r\}$. If $N \cap A_r = 1$, then for $x \in N$ and $y \in A_r$ the commutator $[x, y] = x^{-1}(y^{-1}xy)$ lies in N (as $N \trianglelefteq S_r$) and equals $(x^{-1}y^{-1}x)y \in A_r$ (as $A_r \trianglelefteq S_r$), hence lies in $N \cap A_r = 1$. So N centralises A_r , and $N = 1$ by Lemma 4.1. The quotients are then S_r and $S_r/A_r \cong C_2$.

Finally, in each row of the table the only normal subgroup of index 2 is A_r : for $r = 2$ it is $1 = A_2$; for $r = 3, 4$ and $r \geq 5$ inspection of the lists (whose orders are 1, 3, 6; 1, 4, 12, 24; 1, $r!/2$, $r!$) gives A_r in each case. $\square$

The following consequence is the engine of the whole proof.

Proposition 4.4. *Let $m, n \geq 2$ and let $H \leq \Gamma_{m,n}$ be a subdirect subgroup of $S_m \times S_n$, with common quotient Q as in Lemma 3.1. Then:*

1. Q is nontrivial;
2. if $Q \cong C_2$ then $H = \Gamma_{m,n}$.

Proof. (i) If Q were trivial then, as noted after Lemma 3.1, $H = S_m \times S_n$. But $S_m \times S_n \not\leq \Gamma_{m,n}$: since $m \geq 2$ the transposition $(1\ 2)$ lies in S_m , and the pair $((1\ 2), \text{id})$ has $\text{sgn}((1\ 2)) = -1 \neq +1 = \text{sgn}(\text{id})$, so it lies in $S_m \times S_n$ but not in $\Gamma_{m,n}$. This contradicts $H \leq \Gamma_{m,n}$.

(ii) Suppose $Q \cong C_2$. Then $N_{S_m} \trianglelefteq S_m$ has index 2, so $N_{S_m} = A_m$ by Proposition 4.3; likewise $N_{S_n} = A_n$. The map sgn induces isomorphisms $S_m/A_m \xrightarrow{\sim} \{\pm 1\}$ and $S_n/A_n \xrightarrow{\sim} \{\pm 1\}$, and C_2 has only the identity automorphism, so under these identifications the Goursat isomorphism $\phi : S_m/A_m \rightarrow S_n/A_n$ is the identity of $\{\pm 1\}$; that is, $\phi(\sigma A_m) = \tau A_n$ holds if and only if $\text{sgn}(\sigma) = \text{sgn}(\tau)$. Now (3.1) reads

$$H = \{(\sigma, \tau) : \text{sgn}(\sigma) = \text{sgn}(\tau)\} = \Gamma_{m,n}. \quad \square$$

5 Proof of Theorem 1.1

5.1 Unequal degrees

Proposition 5.1. *Let $m > n \geq 2$ with $(m, n) \neq (4, 3)$. Then $\Gamma_{m,n} = \langle (a_m, a_n), (b_m, b_n) \rangle$.*

Proof. Put $H = \langle (a_m, a_n), (b_m, b_n) \rangle$.

$H \leq \Gamma_{m,n}$. By Lemma 2.2, a_m and a_n are both even and b_m and b_n are both odd, so both generators lie in $\Gamma_{m,n}$, which is a group.

H is *subdirect*. The image of H under the first projection contains a_m and b_m , hence is S_m by Lemma 2.3; symmetrically the second projection is onto S_n .

The common quotient is C_2 . Let Q be the common quotient of H given by Lemma 3.1; it is a quotient of S_m and also of S_n , and it is nontrivial by Proposition 4.4(i). We show $Q \cong C_2$, distinguishing three cases according to m (recall $m > n \geq 2$, so $m \geq 3$).

Case $m \geq 5$. By (4.1), $Q \cong C_2$ or $Q \cong S_m$. In the second case $|Q| = m!$; but Q is a quotient of S_n , so $|Q|$ divides $n!$, whence $m! \leq n!$, contradicting $m > n$. So $Q \cong C_2$.

Case $m = 4$. Then $n \in \{2, 3\}$, and $n = 3$ is excluded by hypothesis, so $n = 2$. By (4.1) the only nontrivial quotient of S_2 is C_2 , so $Q \cong C_2$.

Case $m = 3$. Then $n = 2$, and again $Q \cong C_2$.

By Proposition 4.4(ii), $H = \Gamma_{m,n}$. □

Remark 5.2. The pair $(4, 3)$ must be excluded here for a reason visible in the proof: S_4 and S_3 share the nontrivial quotient S_3 , by Lemma 4.2. This is not merely a gap in the argument — Section 6 shows that the displayed pair really does fail to generate $\Gamma_{4,3}$, and that it fails precisely by landing in the fibre product over S_3 .

5.2 Equal degrees

Proposition 5.3. *Let $r \geq 5$. Then $\Gamma_{r,r} = \langle (a_r, a_r), (b_r, d_r) \rangle$.*

Proof. Put $H = \langle (a_r, a_r), (b_r, d_r) \rangle$.

$H \leq \Gamma_{r,r}$. By Lemma 2.2 a_r is even, so $(a_r, a_r) \in \Gamma_{r,r}$; by Lemma 2.2 b_r is odd, and by Lemma 2.4 d_r is odd as well, so $(b_r, d_r) \in \Gamma_{r,r}$.

H is *subdirect*. The first projection of H contains a_r and b_r , so is S_r by Lemma 2.3. The second projection contains a_r and d_r , so is S_r by Lemma 2.5.

The common quotient is C_2 . Let Q be the common quotient. It is nontrivial by Proposition 4.4(i), and since $r \geq 5$, (4.1) leaves $Q \cong C_2$ or $Q \cong S_r$. Suppose $Q \cong S_r$. Then $|Q| = r! = |S_r|$ forces both Goursat kernels to be trivial, so by the remark following Lemma 3.1 the subgroup H is the graph of an automorphism ϕ of S_r :

$$H = \{(\sigma, \phi(\sigma)) : \sigma \in S_r\}.$$

Both generators of H have this form, so

$$\phi(a_r) = a_r, \quad \phi(b_r) = d_r.$$

But an automorphism preserves the order of every element, while $|b_r| = 2$ and, by Lemma 2.4,

$$|d_r| = \begin{cases} r-1 \geq 4, & r \text{ odd,} \\ r \geq 6, & r \text{ even,} \end{cases}$$

using $r \geq 5$ in both branches. Hence $|b_r| \neq |d_r|$, a contradiction. Therefore $Q \cong C_2$, and Proposition 4.4(ii) gives $H = \Gamma_{r,r}$. □

Remark 5.4. The obstruction just used is an order obstruction, and orders are preserved by *every* automorphism, inner or outer. In particular the argument requires no knowledge of $\text{Aut}(S_r)$; the well-known exceptional outer automorphism of S_6 is immaterial, and the case $r = 6$ needs no separate treatment.

5.3 Conclusion, and the exact value of the rank

Proof of Theorem 1.1. Let $m \geq n \geq 2$ with $(m, n) \notin \{(2, 2), (3, 3), (4, 3), (4, 4)\}$. If $m > n$ then $(m, n) \neq (4, 3)$ and Proposition 5.1 applies. If $m = n = r$ then $r \notin \{2, 3, 4\}$, i.e. $r \geq 5$, and Proposition 5.3 applies. In both cases $\Gamma_{m,n}$ is generated by the two displayed elements, so $\text{rank}(\Gamma_{m,n}) \leq 2$. The reverse inequality is Corollary 5.5 below. $\square$

Corollary 5.5. *For every (m, n) as in Theorem 1.1, the group $\Gamma_{m,n}$ is nonabelian; hence it is not cyclic, $\text{rank}(\Gamma_{m,n}) \geq 2$, and therefore $\text{rank}(\Gamma_{m,n}) = 2$.*

Proof. A group of rank at most 1 is cyclic, hence abelian, so it suffices to prove that $\Gamma_{m,n}$ is nonabelian. Note $m \geq 3$: the only admissible pair with $m = 2$ would be $(2, 2)$, which is excluded.

If $m \geq 4$, then $A_m \times \{\text{id}\} \leq A_m \times A_n \leq \Gamma_{m,n}$ and A_m is nonabelian, since $(1\ 2\ 3)(1\ 2\ 4) = (1\ 3)(2\ 4)$ while $(1\ 2\ 4)(1\ 2\ 3) = (1\ 4)(2\ 3)$.

If $m = 3$, then $n = 2$, because $(3, 3)$ is excluded and $n \leq m$. The first projection $\Gamma_{3,2} \rightarrow S_3$ is surjective (given $\sigma \in S_3$, pair it with the element of S_2 of the same sign) and injective (its kernel is $\{(\text{id}, \tau) : \text{sgn}(\tau) = +1\} = \{(\text{id}, \text{id})\}$), so $\Gamma_{3,2} \cong S_3$, which is nonabelian. $\square$

Together with the known values $\text{rank}(\Gamma_{2,2}) = 1$ and $\text{rank}(\Gamma_{3,3}) = \text{rank}(\Gamma_{4,3}) = \text{rank}(\Gamma_{4,4}) = 3$, Theorem 1.1 determines $\text{rank}(\Gamma_{m,n})$ for all $m \geq n \geq 2$.

6 The exceptional pairs

The exceptional list in Conjecture 1 is exactly the list of pairs at which one of the two mechanisms of Section 5 breaks down, and the breakdown is visible in the constructions themselves; this explains, rather than merely accommodates, the shape of the exceptional set. Orders quoted here were computed by exhaustive closure (Section 7).

$(m, n) = (4, 3)$. The unequal-degree pair of Proposition 5.1 reads $((1\ 2\ 3), (1\ 2\ 3)), ((3\ 4), (1\ 2))$. It generates a subgroup of order 24 inside $\Gamma_{4,3}$, which has order 72; the index is 3. This is exactly the fibre product predicted by Lemma 3.1 over the common quotient S_3 , which is a quotient of S_4 (namely S_4/V_4 , by Lemma 4.2) as well as of S_3 : indeed $|S_4|/|S_3|/|S_3| = 24$.

$(m, n) = (4, 4)$. The equal-degree pair reads $((1\ 2\ 3), (1\ 2\ 3)), ((3\ 4), (1\ 2\ 4\ 3))$; here the order obstruction of Proposition 5.3 is available ($|b_4| = 2 \neq 4 = |d_4|$) and does rule out the quotient S_4 , but the extra quotient $S_4/V_4 \cong S_3$ is not ruled out — and it is precisely what occurs: the generated subgroup has order $96 = 24 \cdot 24/6$, of index 3 in $|\Gamma_{4,4}| = 288$.

$(m, n) = (3, 3)$. Here $|d_3| = |(2\ 3)| = 2 = |b_3|$, so the order obstruction is vacuous, and the graph case really occurs. Conjugation by $a_3 = (1\ 2\ 3)$ is an automorphism ϕ of S_3 with $\phi(a_3) = a_3$ and, by (2.1), $\phi(b_3) = \phi((1\ 2)) = (a_3(1)\ a_3(2)) = (2\ 3) = d_3$. Hence the subgroup generated by (a_3, a_3) and (b_3, d_3) is contained in — and, both projections being onto, equal to — the graph $\{(\sigma, a_3\sigma a_3^{-1})\}$, of order 6 and index 3 in $|\Gamma_{3,3}| = 18$.

$(m, n) = (2, 2)$. $\Gamma_{2,2} \cong C_2$ is cyclic of rank 1, so it is excluded from a conjecture asserting rank exactly 2, though it is of course 2-generated in the weak sense.

In all three nontrivial cases the recipe fails by exactly the index 3 predicted by the corresponding Goursat quotient of order 6. That no *other* choice of two elements succeeds either — the assertion $\text{rank} = 3$ recorded in [1] — we have reconfirmed independently in Section 7.

(m, n)	recipe	$ \langle \cdot \rangle $	$ \Gamma_{m,n} = m!n!/2$	verdict
(3, 2)	unequal	6	6	equal
(4, 2)	unequal	24	24	equal
(5, 2)	unequal	120	120	equal
(5, 3)	unequal	360	360	equal
(5, 4)	unequal	1,440	1,440	equal
(6, 2)	unequal	720	720	equal
(6, 3)	unequal	2,160	2,160	equal
(6, 4)	unequal	8,640	8,640	equal
(6, 5)	unequal	43,200	43,200	equal
(7, 3)	unequal	15,120	15,120	equal
(7, 6)	unequal	1,814,400	1,814,400	equal
(5, 5)	equal	7,200	7,200	equal
(6, 6)	equal	259,200	259,200	equal
(7, 7)	equal	12,700,800	12,700,800	equal
(4, 3)	unequal	24	72	proper, index 3
(3, 3)	equal	6	18	proper, index 3
(4, 4)	equal	96	288	proper, index 3

Table 1: Exhaustive closure of the generating pairs of Theorem 1.1. The first fourteen rows are admissible pairs and the closure is all of $\Gamma_{m,n}$. The last three rows are the nontrivial exceptional pairs, where the recipe demonstrably fails, by the index predicted in Section 6.

7 Computational verification

The proof of Theorem 1.1 is complete and uses no computation. Nevertheless, since the theorem asserts something checkable for each small (m, n) , we verified the explicit generating pairs by exhaustive closure. The computation is a breadth-first search in the Cayley graph: starting from the identity of $S_m \times S_n$ and repeatedly left-multiplying by the two generators, one enumerates the generated subgroup and compares its cardinality with $|\Gamma_{m,n}| = m!n!/2$. Permutations were indexed in lexicographic order and the two generators were pre-tabulated as index maps, so that each edge of the search costs two array lookups. All arithmetic is exact.

Table 1 reports the outcome for fourteen admissible pairs, all those with $m \leq 7$ that we selected, together with the three nontrivial exceptional pairs. In every admissible case the closure has exactly $m!n!/2$ elements, the largest being $|\Gamma_{7,7}| = 12,700,800$; in every exceptional case it is a proper subgroup of index 3. Four further families of checks were run.

1. *Generators.* For $2 \leq r \leq 12$: $\text{sgn}(a_r) = +1$, $\text{sgn}(b_r) = \text{sgn}(d_r) = -1$, $|b_r| = 2$, $|d_r| = r - 1$ for odd r and r for even r , the closed form (2.4) for d_r , and the identity $b_r = d_r a_r^{-1}$: all confirmed. For $2 \leq r \leq 8$, $|\langle a_r, b_r \rangle| = |\langle a_r, d_r \rangle| = r!$: confirmed (Lemmas 2.3, 2.5).
2. *Ranks of the exceptional groups.* By exhaustive sweep over all ordered pairs of elements: $\Gamma_{2,2}$ (order 2) is cyclic; and for $\Gamma_{3,3}$ (order 18), $\Gamma_{4,3}$ (order 72) and $\Gamma_{4,4}$ (order 288) *no* pair of elements generates. This independently reconfirms the values $\text{rank} = 3$ quoted from [1] and used in Section 6, and it is what makes the exceptional list genuinely exceptional rather than an artefact of our particular recipe.
3. *Structural lemmas.* The conjugacy-class sizes of S_4 were recomputed (1, 6, 3, 8, 6 for types id, 2, 2+2, 3, 4), confirming the enumeration in Proposition 4.3; the homomorphism $\rho : S_4 \rightarrow \text{Sym}(\Pi)$ of Lemma 4.2 was constructed explicitly and found to have image of order 6, nonabelian, and kernel exactly V_4 ; and the centraliser of A_r in S_r was computed for

$3 \leq r \leq 6$, giving order 3 for $r = 3$ and order 1 for $r = 4, 5, 6$, confirming both Lemma 4.1 and the necessity of its hypothesis $r \geq 4$.

4. *The isomorphism $\Gamma_{3,2} \cong S_3$ used in Corollary 5.5 was confirmed by checking that the first projection is injective on $\Gamma_{3,2}$, which has order 6.*

None of these computations is load-bearing for Theorem 1.1; all of them agree with it.

8 The verification chain

We describe how this result was produced and checked, because the answer bears on how much weight a reader should give it before checking it personally.

Solver/verifier separation. The proof was found by OpenAI GPT-5.6 running in the `codex` command-line interface. The problem was presented in disguised form: the task file stated the group $\Gamma_{m,n}$, the exceptional list and the known ranks, but named neither Fernandes nor the arXiv identifier, and the solver was instructed not to search the internet and not to read any other file in the workspace. It was required to present only what it could rigorously prove, and was explicitly permitted to return a partial answer; the solution it returned covered every admissible pair and was correct.

Line-by-line verification. The argument was then checked step by step by Claude (Table 5), including independent recomputation of the cycle forms, parities and orders of a_r, b_r, d_r , of the conjugacy-class enumeration for S_4 , and of the closures of Section 7.

Adversarial review. The argument was submitted, in two separate sessions with no access to the solver’s reasoning trace, to two reviewers, under a protocol that requires each objection to be classified as a *critical error* (breaking the logical chain) or a *justification gap* (a true step insufficiently argued), and requires a final verdict. One reviewer was Qwen3.8-Max, from a different vendor than the solver; the other was GPT-5.6 at “Pro” effort, which is the same vendor as the solver in a different harness and at a different effort setting, and its independence should therefore be discounted accordingly.

GPT-5.6 returned `VALID`, with no issues, having independently confirmed the cycle form of $b_r a_r$ under right-to-left composition and the resulting orders $r - 1$ and r , the generation statement including the edge case $r = 2$, the normal-quotient classification, the exclusion of quotients other than C_2 in the unequal case, the surjectivity of the second projection via $b_r = d_r a_r^{-1}$, and the order argument against the graph case including $r = 6$.

Qwen3.8-Max returned `INCOMPLETE`: no critical errors, but six justification gaps, all of them about what was asserted rather than proved. We list each objection with the place in this write-up that answers it; we agree with the reviewer that all six were gaps in exposition and none was a gap in the mathematics.

1. $\Gamma_{m,n}$ was used before being defined. Answered by (1.1).
2. *The well-definedness, bijectivity and converse inclusion in the Goursat lemma were asserted, not proved.* Answered by the complete proof of Lemma 3.1.
3. $S_4/V_4 \cong S_3$ was asserted, whereas the class-size argument bounds only the order of the quotient, leaving C_6 formally open. Answered by Lemma 4.2, which exhibits the action of S_4 on the three pair-partitions of $\{1, 2, 3, 4\}$ and identifies the kernel, and thereby also shows the quotient nonabelian.
4. *The triviality of the centraliser of A_r in S_r was used without proof.* Answered by Lemma 4.1, which also records that its hypothesis $r \geq 4$ cannot be dropped; it is applied only for $r \geq 5$,

the cases $r \leq 4$ of Proposition 4.3 being handled by direct enumeration.

5. *The parity of d_r was not derived.* Answered by Lemma 2.4, which computes the cycle form of d_r and reads the parity off the (even) cycle length in both branches.
6. *The step “a trivial common quotient would make $H = S_m \times S_n$, which is not contained in $\Gamma_{m,n}$ ” did not say why the containment fails.* Answered by the explicit witness $((1\ 2), \text{id})$ in the proof of Proposition 4.4(i).

Computational verification. As reported in Section 7.

Formal verification. The main theorem is now formally verified. The Lean 4 file `Fernandes.lean` (507 lines) compiles with zero errors and zero warnings, contains no `sorry` and no `native_decide`, and introduces no new axioms: `#print axioms` on each of its five public theorems reports exactly `[propxt, Classical.choice, Quot.sound]`, the three axioms in routine use throughout `mathlib`. The statement of the main theorem, `conjecture_1` — including the auxiliary definitions `signDiffHom` and `gammaSubgroup` — is byte-identical to the statement of `conjecture_1` in `FormalConjectures/Arxiv/2605.12342/Conjecture1.lean` [5]; only the proof body differs. We also formalised the variant `conjecture_1.variants.rank_2_2` ($\Gamma_{2,2}$ is cyclic), discharging a second of the five `sorry` placeholders in that file. The remaining three variants, recording that $\Gamma_{3,3}$, $\Gamma_{4,3}$ and $\Gamma_{4,4}$ have rank 3, are not formalised; their verification remains the computational argument of Section 7.

Formalisation notes. The formal proof is not a line-by-line translation of Sections 2–5, but an independently verified route to the same statement, which is arguably additional evidence for the theorem rather than a mechanical transcription of the write-up above. Four simplifications are worth recording. First, `mathlib`’s `Equiv.Perm.alternatingGroup_le_of_normal` (for $r \geq 5$) replaces Lemma 4.1 and Proposition 4.3 entirely. Second, the formalisation uses a single generating pair uniformly in r — $c = \text{finRotate } r$ and $t = \text{swap } 0\ 1$ — whose roles under the sign character swap automatically with the parity of r , so the even/odd case split of Section 2 never appears. Third, the Goursat step is carried out with kernels alone (`goursatFst`, `goursatSnd`); the quotient isomorphism ϕ of Lemma 3.1 is never constructed. Fourth, the order obstruction of Proposition 5.3 is weakened, and suffices in this weaker form, to $d^2 \neq 1$.

Prior work. Kenta Kitamura (GitHub user `KitaKen1`) has an earlier, independent kernel-checked Lean 4 proof of the exact `formal-conjectures` statement of Conjecture 1, linked from `google-deepmind/formal-conjectures` pull request #4868 (“mark Fernandes conjecture 1 as solved”), opened 11 August 2026 and, as of this writing (18 August 2026), still open and unmerged. That date precedes the start of the present project (16 August 2026), so Kitamura’s formalization has priority; our Lean development described above and the elementary argument of Sections 2–5 were both found independently, by a different, unrelated pipeline, without knowledge of his proof. We record our proof anyway because we believe the human-readable argument here — via Goursat’s lemma and the classification of normal subgroups of S_r — may retain expository value alongside the formal one, even though it establishes nothing that was not already machine-checked.

9 Limitations

1. *The proof is elementary.* It combines two textbook ingredients — Goursat’s lemma and the classification of normal subgroups of S_r — with one small design choice, namely replacing the symmetric second generator (b_r, b_r) by (b_r, d_r) so that the two components have different orders. There is no new technique here. The value of the result is that it settles a conjecture stated in 2026 and open at the time of writing, not that the method is novel.

2. *The equal-degree construction is not canonical.* Many other second generators would work; ours is chosen to make the order obstruction as short as possible. We have not investigated which pairs of elements generate $\Gamma_{r,r}$ in general, nor the proportion of generating pairs (a natural question in the tradition of P. Hall’s Eulerian functions [4]).
3. *Literature work was automated, and we did not consult the source article.* Our statement of Conjecture 1, of the known ranks of the four exceptional pairs, and of the scope of Fernandes’ Corollary 4, is taken from the Lean transcription in **formal-conjectures** [5] and from an automated literature pass, not from the arXiv PDF; the same applies to [2], cited only as background. Theorem 1.1 and its proof are self-contained and independent of all of that, but attributions — in particular our characterisation of what Corollary 4 of [1] covers — should be read with the caveat. An initial automated search for a prior proof of Conjecture 1 found none; a later, more targeted search did locate one — K. Kitamura’s independent Lean 4 formalization, predating this project (Section 8) — so we no longer treat the earlier negative search as informative, and any remaining priority claim on our part should be discounted accordingly.
4. *Formalisation is complete for the main theorem and one variant.* The Lean 4 formalisation (Section 8) proves Theorem 1.1 against the **formal-conjectures** statement of Conjecture 1, together with the variant recording $\text{rank}(\Gamma_{2,2}) = 1$. The three remaining variants, for the exceptional ranks $\text{rank}(\Gamma_{3,3}) = \text{rank}(\Gamma_{4,3}) = \text{rank}(\Gamma_{4,4}) = 3$, are not formalised; their verification remains the exhaustive computation of Section 7, which is a complete proof but not a machine-checked one in the Lean sense.
5. *The exceptional cases are quoted, not proved here.* We reconfirmed $\text{rank}(\Gamma_{3,3}) = \text{rank}(\Gamma_{4,3}) = \text{rank}(\Gamma_{4,4}) = 3$ by exhaustive computation, which is a complete proof for those three finite groups, but a computational one; we give no structural explanation of why no pair generates. Conjecture 1 as stated presupposes those values, though Theorem 1.1 does not depend on them: the assertion $\text{rank}(\Gamma_{m,n}) = 2$ for admissible (m, n) is proved in Corollary 5.5 without reference to them.

Acknowledgement of process

No human contributed a mathematical step to this paper. The pipeline — problem selection, disguised task construction, solving, verification, adversarial review, computation, and drafting — was run autonomously. Responsibility for the correctness of the argument nevertheless rests where it always does: with the reader who checks it.

References

- [1] V. H. Fernandes, *Groups of permutations that are even on maximal proper subsets, and related monoids*, arXiv:2605.12342 (2026). Conjecture 1 is the statement proved here; Corollary 4 of that paper establishes a special case.
- [2] V. H. Fernandes and A. Vernitski, *Internat. J. Algebra Comput.* (2026). Cited as background on the ranks of the monoids in whose study $\Gamma_{m,n}$ arises. Bibliographic details are as supplied by our automated literature pass; we did not consult the article (see Section 9).
- [3] É. Goursat, *Sur les substitutions orthogonales et les divisions régulières de l’espace*, Ann. Sci. École Norm. Sup. (3) **6** (1889), 9–102. The source of Lemma 3.1.

- [4] P. Hall, *The Eulerian functions of a group*, Quart. J. Math. Oxford Ser. **7** (1936), 134–151.
- [5] Google DeepMind and contributors, *formal-conjectures*: a corpus of open conjectures stated in Lean 4, <https://github.com/google-deepmind/formal-conjectures>. The statement proved here was added as `FormalConjectures/Arxiv/2605.12342/Conjecture1.lean` on 9 August 2026 (Issue #4815, PR #4836), with five `sorry` placeholders: the conjecture itself and four variants recording the ranks of the exceptional pairs.
- [6] The mathlib Community, *The Lean mathematical library*, Proc. 9th ACM SIGPLAN Int. Conf. on Certified Programs and Proofs (CPP 2020), 367–381.